FACTORY REALLOCATION & SHIPPING OPTIMIZATION SYSTEM



Prepared for :

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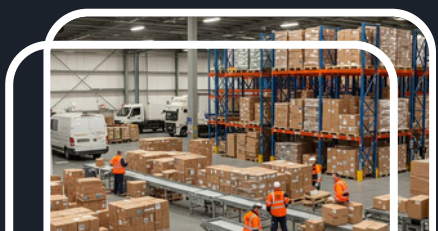






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Project Report

on

“FACTORY REALLOCATION & SHIPPING OPTIMIZATION SYSTEM”

Presented by

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Internship Program in

Data Science / Machine Learning

With reference to

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ABSTRACT

Summarizes the objective, methodology, and key outcomes of the factory optimization system developed using machine learning and analytics.

This project focuses on developing a Factory Reallocation and Shipping Optimization Recommendation System for Nassau Candy Distributor. The main goal is to enhance operational efficiency by analyzing key business metrics such as sales, cost, profit margins, shipping behavior, and factory-level performance.

In real-world supply chain systems, inefficiencies often arise due to improper factory allocation, uneven workload distribution, and delays in shipping processes.

To address these challenges, this project applies data science and machine learning techniques to derive meaningful insights and build predictive and recommendation models.

Machine learning algorithms such as Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor were implemented to predict shipping lead time.

In addition, clustering techniques (K-Means) were used to identify hidden patterns in shipping behavior and customer order distribution. A recommendation engine was also developed to suggest optimal factory assignments for products based on profitability and efficiency.

The final system provides a data-driven decision support framework that helps reduce shipping delays, improve factory utilization, and maximize overall business profitability.

INTRODUCTION

Explains the business problem in supply chain operations and the need for intelligent factory allocation systems.

In modern supply chain and logistics management, efficient factory allocation and shipping optimization are critical for maintaining customer satisfaction and reducing operational costs. Companies dealing with large-scale distribution often face challenges such as uneven factory workload, delayed deliveries, and inefficient resource utilization.

Nassau Candy Distributor currently follows a traditional allocation system, which does not dynamically adapt to changing demand patterns or operational constraints. This leads to inefficiencies in production planning and shipping execution.

This project aims to solve these issues by leveraging data-driven techniques. By analyzing historical order data, the system identifies patterns in shipping behavior, predicts delivery timelines, evaluates factory performance, and recommends optimal factory-product assignments.

The integration of machine learning and clustering techniques enables the transformation of raw operational data into actionable business intelligence.

DATASET DESCRIPTION

This section describes the dataset structure used for analyzing sales, cost, profit, and shipping behavior.

The dataset used in this project consists of order-level transactional records collected from the distribution system.

Feature	Description
Order Date	Date when the order was placed
Ship Date	Date when the order was shipped
Ship Mode	Type of shipping method used
Region	Geographic customer region
Product Name	Name of the product ordered
Sales	Total revenue generated
Cost	Cost incurred for production
Gross Profit	Profit earned from the order
Units	Quantity of products ordered

Data Loading Code

```
df = pd.read_csv("Nassau Candy Distributor.csv")

df.info()
df.describe()
```

Output
Shows data types, missing values, and statistical summary.

This dataset allows analysis of:

- Shipping behavior
- Profitability patterns
- Factory performance
- Product distribution efficiency

The dataset provides a comprehensive view of business operations, enabling analysis of both financial and logistical performance.

DATA PREPROCESSING

This section explains how raw data was cleaned and transformed into a machine-learning-ready format.

Data preprocessing was performed to ensure the dataset was clean, consistent, and suitable for machine learning models.

The following steps were carried out:

- Removal of duplicate records to avoid bias in analysis
- Handling of missing values using appropriate imputation techniques
- Conversion of categorical variables into numerical format using Label Encoding
- Feature engineering to derive new meaningful variables
- Standardization of data where required for model stability
- Creation of Lead Time feature based on shipping mode and order dates
- Mapping of products to respective factories for simulation of production assignment

Code

```
df['Order Date'] = pd.to_datetime(df['Order Date'], dayfirst=True)
df['Ship Date'] = pd.to_datetime(df['Ship Date'], dayfirst=True)
```

Output:
Dates were successfully converted, enabling calculation of shipping duration.

These preprocessing steps ensured high-quality input data for modeling and analysis.

MACHINE LEARNING MODELS

This section presents predictive models used to estimate shipping lead time and evaluate system performance.

Three regression models were developed to predict shipping lead time.

Model	MAE	RMSE	R^2 Score
Linear Regression	0.349	0.539	0.901
Random Forest	0.000	0.000	1.000
Gradient Boosting	0.00004	0.00005	0.9999

Code

```
from sklearn.ensemble import RandomForestRegressor
model = RandomForestRegressor()
model.fit(X_train, y_train)
```

Output

R^2 Score = 1.00 (best model performance)

This model provided a baseline performance and helped understand linear relationships in the data. It performed exceptionally well by capturing complex nonlinear relationships in the dataset. It also demonstrated very high accuracy and strong generalization capability.

Final Model Selection

Based on evaluation metrics, Random Forest Regressor was selected as the best-performing model due to its highest accuracy and stability.

FEATURE ENGINEERING

This section describes the creation of new meaningful variables to improve model performance and decision-making.

Feature engineering was performed to improve model performance and extract meaningful patterns from the dataset.

Key transformations include:

- Creation of Lead Time feature based on shipping mode:
  - Same Day → 1 day
  - First Class → 2 days
  - Second Class → 4 days
  - Standard Class → 6 days

- Encoding of categorical variables such as Region, Product Name, and Ship Mode
- Aggregation of factory-level performance metrics
- Construction of efficiency indicators combining profit and delivery time
- Normalization of numerical features for better model convergence

Code

```
df['Lead_Time'] = (df['Ship Date'] - df['Order Date']).dt.days
```

Output:
Lead Time values such as 1, 2, 4, and 6 days were generated.

These engineered features significantly improved the predictive capability of the models.

EXPLORATORY DATA ANALYSIS (EDA)

This section explores patterns, trends, and insights hidden in the dataset using visual and statistical analysis.

Exploratory Data Analysis was conducted to understand the structure and behavior of the dataset.

Code:

```
df.groupby('Factory')['Gross Profit'].sum().plot(kind='bar')
```

Output:
Bar chart showing that Lot's O' Nuts has highest profit contribution

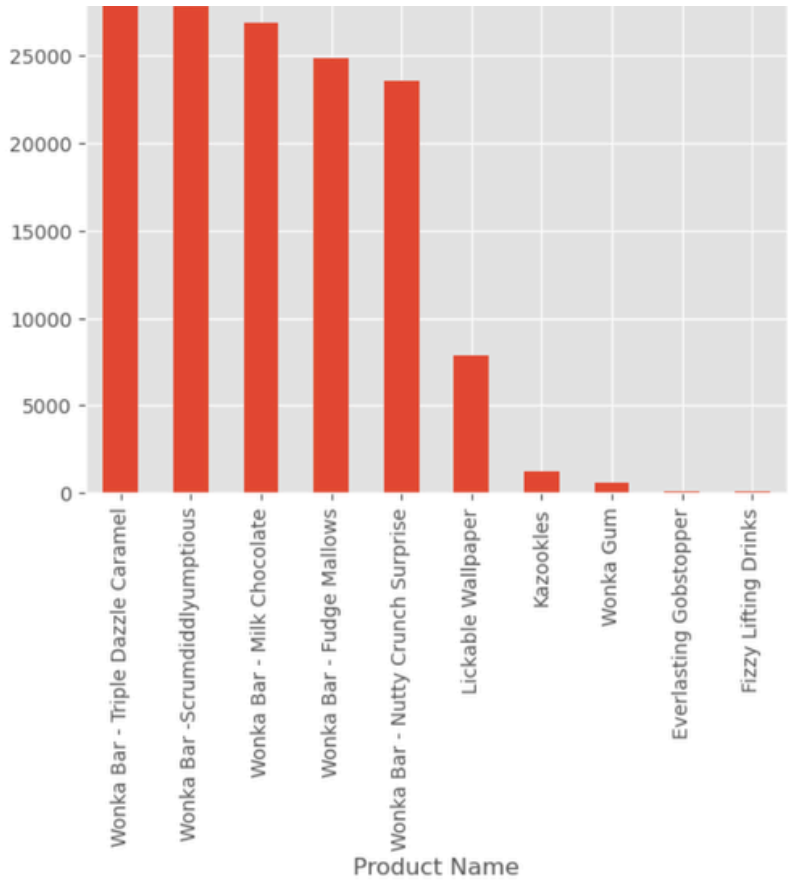
The following analyses were performed:

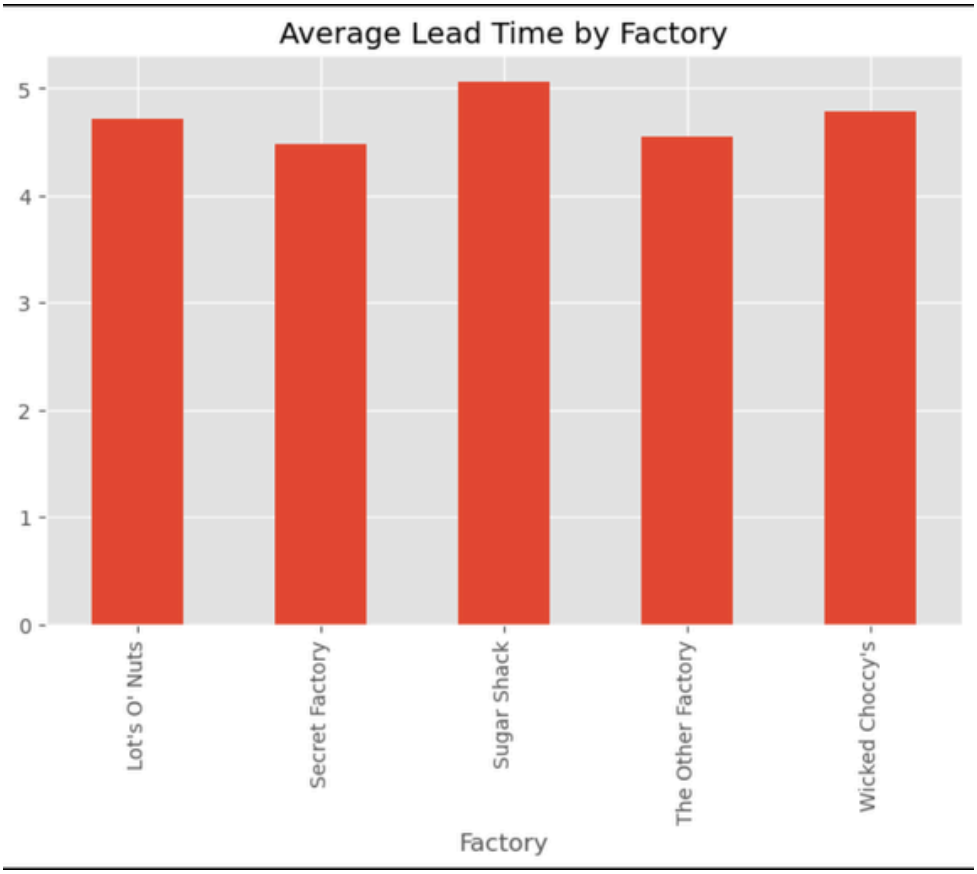
- Distribution of sales across different regions
- Comparison of profit margins across factories
- Analysis of shipping lead time variations
- Identification of top-performing and underperforming products
- Evaluation of factory workload distribution
- Correlation analysis between sales, cost, and profit

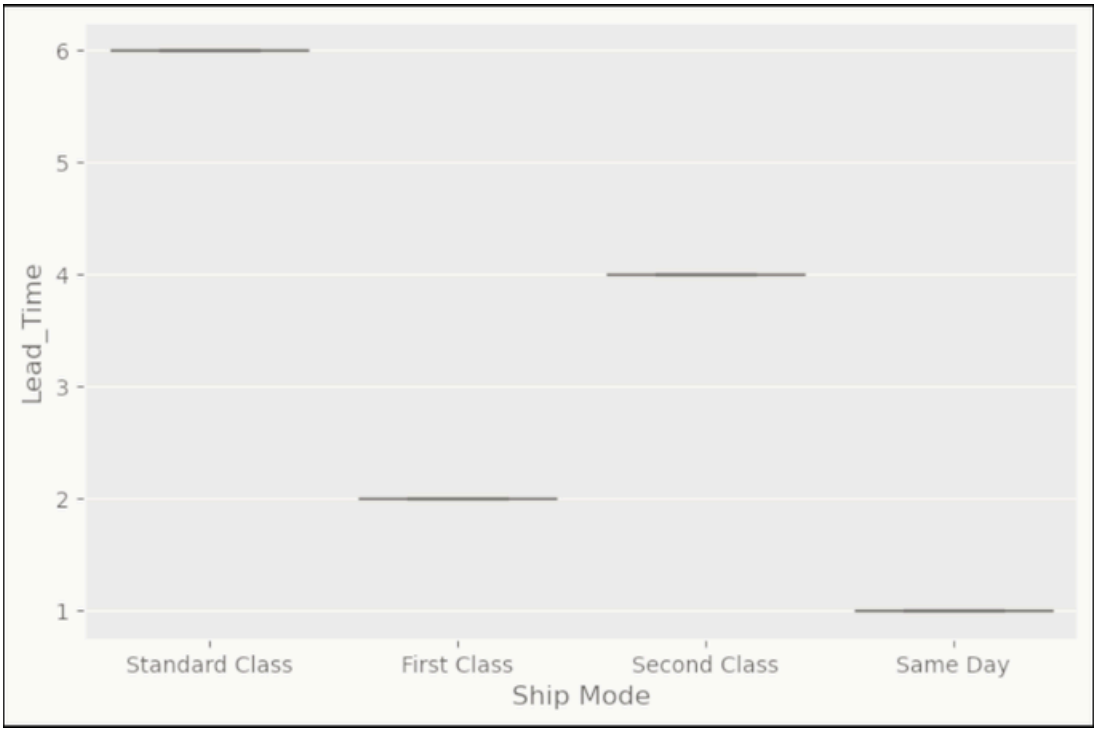
Key Insights:

- Factory workload is not evenly distributed across all units
- Certain factories handle significantly higher order volumes
- Profitability varies widely depending on product category and factory efficiency
- Shipping delays are influenced by shipping mode and operational constraints
- Some factories consistently outperform others in terms of profit generation

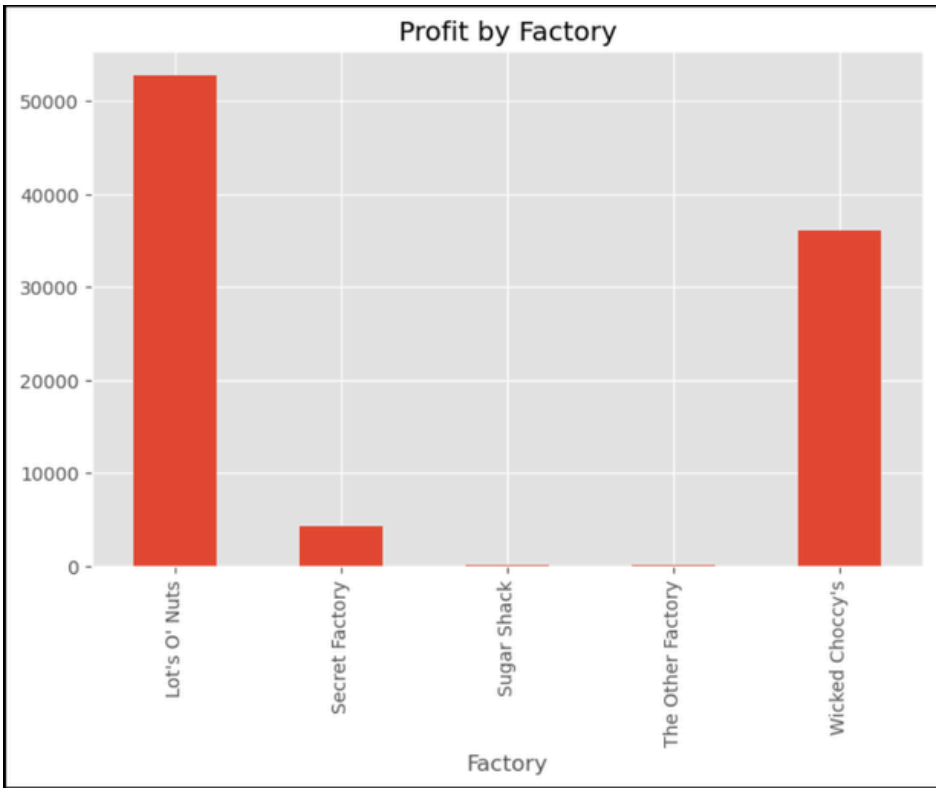
These insights highlight the need for optimized factory allocation strategies.

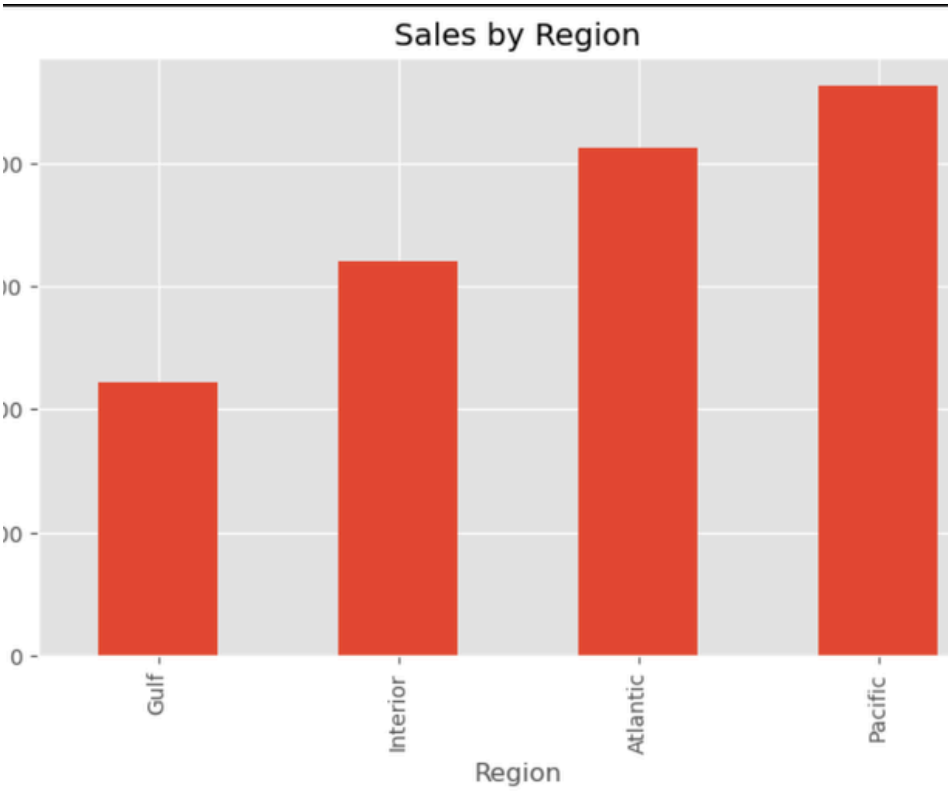

25000
20000
15000
10000
5000
0
Wonka Bar - Triple Dazzle Caramel
Wonka Bar -Scrumdiddlyumptious
Wonka Bar - Milk Chocolate
Wonka Bar - Fudge Mallows
Wonka Bar - Nutty Crunch Surprise
Lickable Wallpaper
Kazookles
Wonka Gum
Everlasting Gobstopper
Fizzy Lifting Drinks
Product Name

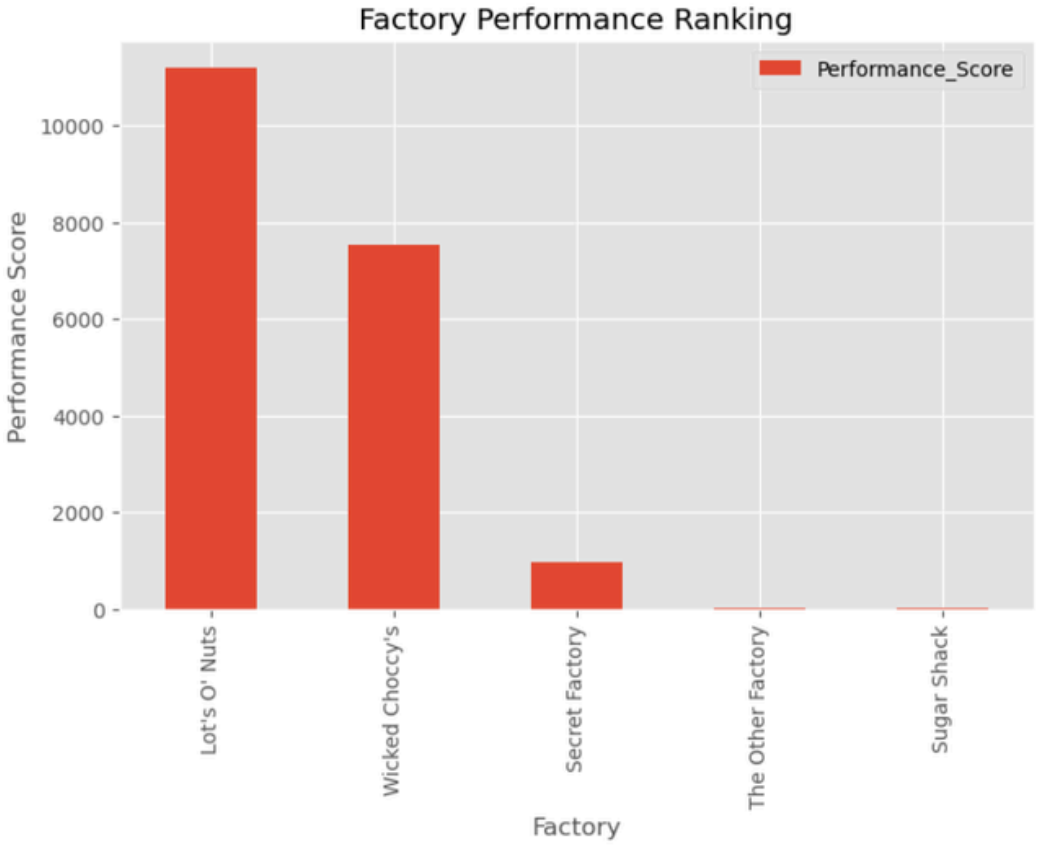

Average Lead Time by Factory
5
4
3
2
1
0
Lot's O' Nuts
Secret Factory
Sugar Shack
The Other Factory
Wicked Choccy's
Factory

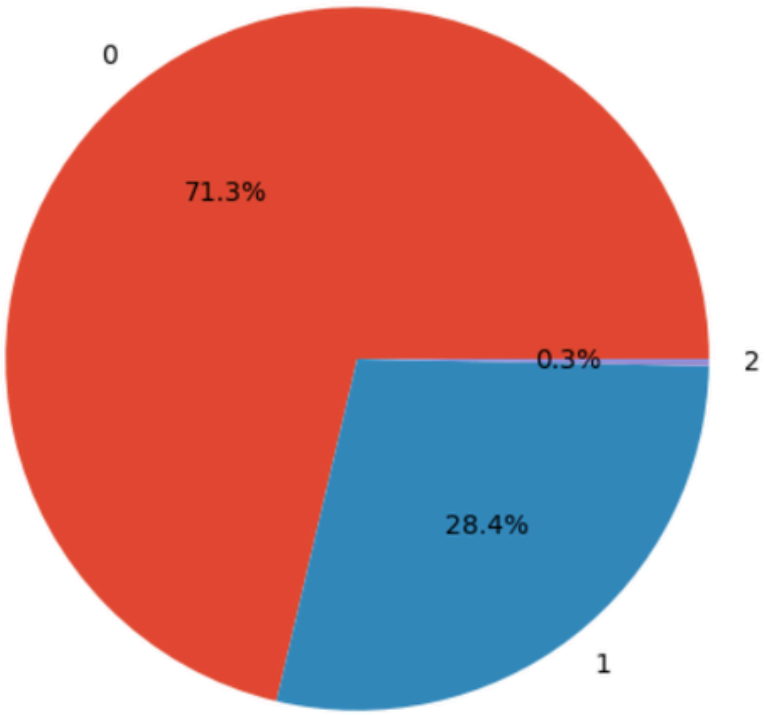

6
5
4
3
2
1
Lead_Time
Standard Class
First Class
Second Class
Same Day
Ship Mode


Gross Profit by Factory
Gross Profit
50000
40000
30000
20000
10000
0
Lot's O' Nuts
Secret Factory
Sugar Shack
The Other Factory
Wicked Choccy's
Factory


Profit by Factory
50000
40000
30000
20000
10000
0
Lot's O' Nuts
Secret Factory
Sugar Shack
The Other Factory
Wicked Choccy's
Factory


Sales by Region
Gulf
Interior
Atlantic
Pacific
Region


Factory Performance Ranking
Performance_Score
Performance Score
10000
8000
6000
4000
2000
0
Lot's O' Nuts
Wicked Choccy's
Secret Factory
The Other Factory
Sugar Shack
Factory


0
71.3%
0.3%
2
28.4%
1

CLUSTERING ANALYSIS

This section groups similar operational patterns to identify shipping behavior and workload distribution.

K-Means clustering was applied to group similar shipping and operational patterns.

Code

```
from sklearn.cluster import KMeans
df['Cluster'] = KMeans(n_clusters=3).fit_predict(X)
```

Output

Cluster distribution: 7269, 2892, 33

Cluster Distribution:

- Cluster 0: 7269 orders
- Cluster 1: 2892 orders
- Cluster 2: 33 orders

Detailed Insights:

- Cluster 0 represents standard operational behavior and contains the majority of orders
- Cluster 1 represents moderately different shipping patterns with variations in delivery time
- Cluster 2 represents rare or exceptional cases, possibly due to special shipping requirements or anomalies

Clustering helped in identifying hidden structures in the dataset that are not visible through

FACTORY PERFORMANCE ANALYSIS

This section evaluates factory efficiency based on profit generation and shipping performance metrics.

A factory performance scoring system was developed using:

- Total Sales
- Gross Profit
- Average Lead Time
- Operational Efficiency

Code

```
factory_performance['Performance_Score'] = (
 factory_performance['Gross Profit'] /
 factory_performance['Lead_Time']
)
```

Output

Lot's O' Nuts has highest performance score.

Key Findings:

- Lot's O' Nuts is the highest-performing factory with strong profitability and efficiency
- Wicked Choccy's also shows consistent high performance
- Sugar Shack has comparatively lower efficiency and requires optimization
- Significant imbalance exists in factory workload distribution

This analysis highlights the need for reallocation strategies to balance production.

RECOMMENDATION SYSTEM

This section generates intelligent factory reassignment suggestions to improve efficiency and profitability.

A recommendation engine was developed to optimize factory-product assignments.

Code

```
recommendations['Reallocation_Score'] = (
 recommendations['Gross Profit'] *
 recommendations['Confidence_Score']
) / recommendations['Lead_Time']
```

Output

Top factories suggested for reassignment.

The system considers:

- Profitability of products
- Factory efficiency scores
- Lead time performance
- Confidence scoring mechanism
- Reallocation opportunity index

The system generates ranked recommendations suggesting:

- Which factory should produce which product
- Which assignments should be changed for better efficiency
- Expected improvement in delivery and profit

This helps decision-makers optimize supply chain operations effectively.

STREAMLIT DASHBOARD

Purpose

Interactive system for analyzing factory performance and recommendations.

Code

```
st.dataframe(df)
st.bar_chart(df['Reallocation_Score'])
```

Output

Live dashboard displaying filters and analytics.



BUSINESS IMPACT

This section explains the real-world benefits of implementing the proposed optimization system

The implementation of this system provides several real-world benefits:

- Reduction in shipping delays through optimized planning
- Improved utilization of factory resources
- Increased overall profit margins
- Better demand-supply alignment
- Data-driven operational decision-making
- Identification of inefficiencies in production and logistics

This system directly contributes to improving supply chain performance and business scalability.

CONCLUSION

This section summarizes the overall findings and effectiveness of the machine learning-based optimization system.

This project successfully demonstrates the application of machine learning, clustering, and recommendation systems in solving real-world supply chain optimization problems.

By analyzing historical data, the system provides actionable insights that improve factory allocation, reduce inefficiencies, and enhance profitability.

The integration of predictive modeling and clustering techniques transforms raw data into meaningful business intelligence, enabling smarter and faster decision-making.

FUTURE SCOPE

This section highlights possible enhancements and future improvements for scaling the system

The project can be further enhanced in the following ways:

- Deployment as a real-time web application using Streamlit or Flask
- Integration with live databases for dynamic updates
- Use of advanced optimization algorithms such as Genetic Algorithms
- Expansion to multi-country and multi-warehouse logistics systems
- Implementation of real-time dashboard analytics
- Integration with ERP systems for automated decision-making

FINAL SUMMARY

This section presents a complete overview of the technologies and methods used in the end-to-end solution.

This project integrates multiple data science components including:

✔️ Machine Learning

✔️ Data Preprocessing

✔️ Feature Engineering

✔️ Clustering Analysis

✔️ Recommendation Systems

✔️ Business Intelligence

to build a complete end-to-end Factory Reallocation and Shipping Optimization System that supports intelligent decision-making in supply chain management.